Errors

Hypothesis Testing

A systematic way to **make decisions** about the conclusions we can draw from our data.

Types of Errors:

Type I Error	False Positive	Incorrectly reject a null hypothesis that is actually true
Type II Error	False Negative	Incorrect fail to reject a null hypothesis that is actually false

Choosing a Significance Level

P-Value:
Probability of a Type I Error if we were to reject the null hypothesis.
Reject the null hypothesis when the p-value is below the significance level.

Decisions about the significance level should be based on the “cost” of making a Type I Error.

Setting Up Tests

Statistical Power: Probability of NOT making a Type II Error.

- Statistical power increases with sample size.
- Parametric tests are generally more powerful than non-parametric ones.

Opt for tests that are more powerful.

Demo

Reviewing test results (chi squared test)

Knowledge Check

Suppose that penguin species and sex were not independent (i.e., not randomly sampled). What type of error would we be making with our statistical test?

a) **Type I Error**
b) **Type II Error**

Knowledge Check

Suppose that male and female gentoo penguins actually do not have different flipper sizes. What type of error would we be making with our statistical test?

a) Type I Error
b) Type II Error

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